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MERCHANT RISK MONITORING

Transaction Risk Analysis & Dashboard



Project Type: Data Analytics Portfolio

Tools: Python | SQL | Power BI

PROJECT OVERVIEW

Project Background

This project focuses on monitoring transaction performance and identifying potential risk patterns across merchants. The analysis examines failed transactions, refund activities, payment methods, transaction amounts, and time-based patterns to identify merchants that may require further investigation.

Project Objectives

- Monitor overall transaction performance.
- Identify merchants with unusually high failure rates.
- Analyze transaction patterns across payment methods and time periods.
- Provide insights to support risk monitoring and further investigation.

Project Scope

- 10,000 transactions
- 50 merchants
- 4 payment methods
- Analysis period: October–November 2026

Data Source: Synthetic Data (Simulated Transaction Data)



DATA OVERVIEW

Dataset Summary

This project uses simulated transaction data to evaluate merchant risk and transaction performance.

Transaction Dataset

- 10,000 transaction records
- 50 unique merchants
- Transaction date and time
- Transaction amount
- Payment method
- Transaction status
- Refund indicator

Merchant Dataset

- 50 merchant records
- Merchant ID and name
- Business category
- Registration date
- Account status

Data Fields

Category	Key Fields
• Transaction	• Transaction ID, Merchant ID, Customer ID
• Transaction Details	• Date & Time, Amount, Payment Method
• Transaction Outcome	• Status, Refund Indicator

Data Period: October–November 2026

Data Source: Synthetic Data (Simulated)

10,000	497	4.97%	2.87%
TOTAL TRANSACTIONS	FAILED TRANSACTIONS	FAILED RATE	REFUND RATE

RANSACTION RISK ANALYSIS

Key Findings

- A total of 10,000 transactions were analyzed across 50 merchants.
- 497 transactions failed, representing an overall Failed Rate of 4.97%.
- The overall Refund Rate was 2.87%.
- Transaction performance varied across merchants and payment methods.

Initial Observation

These results provide an overview of transaction performance and help identify merchants and payment channels that may require further investigation.

Note: Failed transactions and refunds are monitoring indicators, not direct evidence of fraudulent activity.

MERCHANT RISK INVESTIGATION

Merchant M048 recorded the highest Failed Rate among the 50 merchants, with 16 failed transactions out of 167 total transactions.

Its Failed Rate of 9.58% was approximately 1.93 times the overall system average of 4.97%

M048
MERCHANT ID

167
TOTAL TRANSACTIONS

16
FAILED TRANSACTIONS

9.58%
FAILED RATE

Payment Method Analysis

- Debit Card: 16.22% Failed Rate
- Credit Card: 11.90% Failed Rate
- QR Payment: 6.98% Failed Rate
- E-Wallet: 4.44% Failed Rate

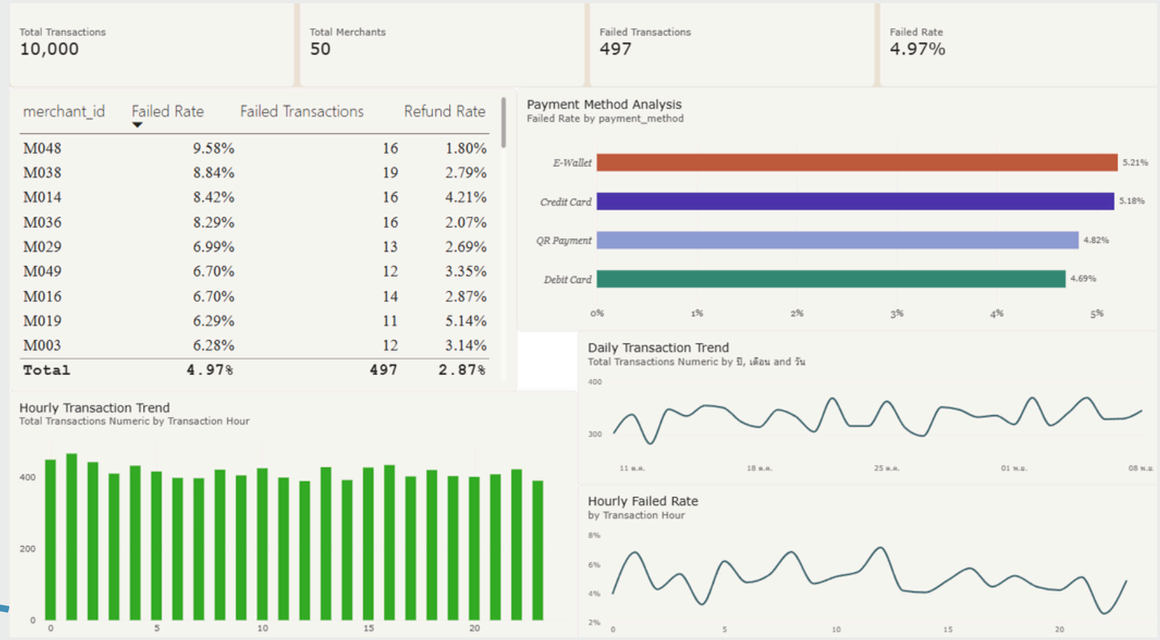
Key Observation

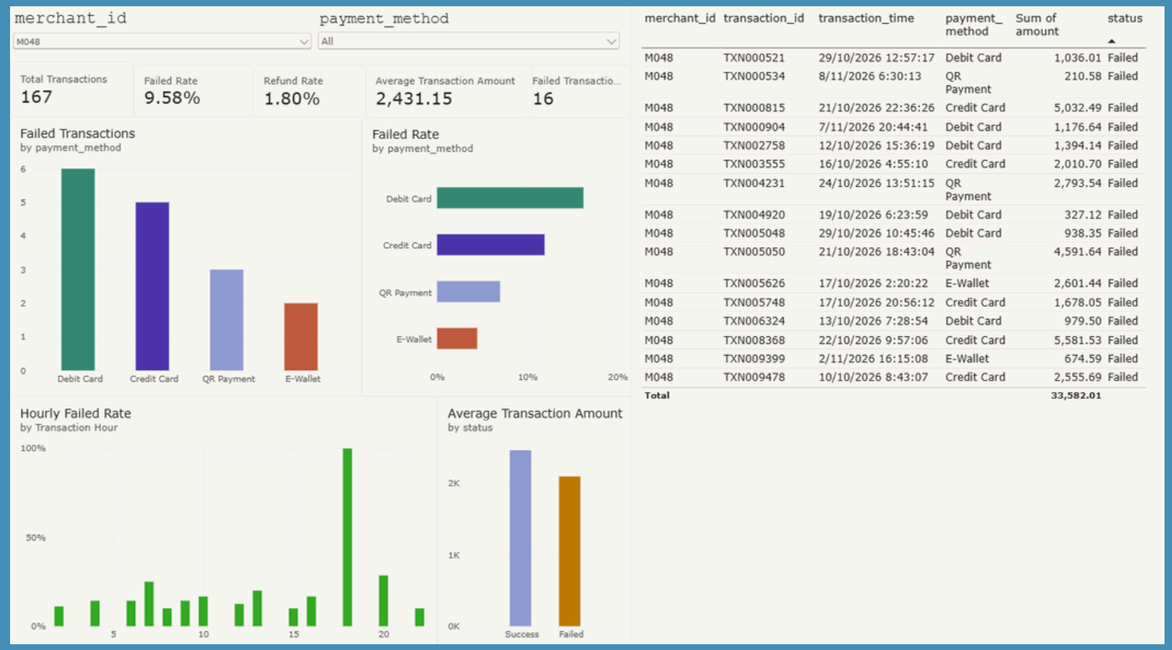
Debit Card and Credit Card transactions showed notably higher failure rates than their respective system averages.

Investigation Priority

Further investigation should focus on payment processing performance, particularly Debit Card and Credit Card transactions.

DASHBOARD & KEY FINDINGS





Key Findings

- Identified Merchant M048 as the merchant with the highest Failed Rate (9.58%).
- Found elevated failure rates in M048's Debit Card (16.22%) and Credit Card (11.90%) transactions compared with system averages.
- Observed hourly variations in transaction failures, while considering transaction volume when interpreting the results.
- Developed an interactive dashboard to support merchant monitoring and risk investigation.

Dashboard Tool: Microsoft Power BI

RECOMMENDATIONS & CONCLUSION

Prioritize Merchant Monitoring

Monitor Merchant M048 closely due to its highest Failed Rate of 9.58% among the 50 merchants.

Investigate Payment Processing

Review Debit Card and Credit Card transactions for M048, as both showed higher Failed Rates than their system averages.

Examine Failure Reasons

Investigate transaction decline codes, payment gateway responses, and issuer-related issues to identify potential causes.

Continue Risk Monitoring

Track Failed Rates and transaction patterns over time to detect unusual changes and support timely investigation.

Conclusion

This project demonstrates how Python, SQL, and Power BI can be used to analyze transaction data, identify potential risk indicators, and develop an interactive merchant monitoring dashboard.
The findings provide a starting point for further investigation rather than definitive evidence of fraud.